

Ethan Tran

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EDUCATION

California State University Sacramento
Bachelor of Science, Computer Science

Expected December 2027
GPA: 3.47

WORK EXPERIENCE

Software Engineer Intern
Perio Specialist

July 2026 - Present

- Developed a machine learning periodontal bone-loss assessment model using pose estimation to identify clinically relevant landmarks in dental radiographs.
- Architected a reliable database for tracking dental implant inventory, usage, and restocking across multiple workstations.
- Established data standardization tooling for converting patient radiographs into model-ready features, optimizing dataset creation.
- Implemented post-processing aligning predicted landmarks with tooth boundaries, improving fine-grained boundary accuracy by 4%.
- Built offline syncing, data validation, backups, and observability to prevent lost or duplicate records.

Pfizer AI Engineering Externship
Extern

June 2026 - July 2026

- Built an OCR-backed document pipeline to extract, classify, and route content from 1,000+ page pharmaceutical files.
- Partnered with Pfizer host to translate document-routing pain points into pipeline requirements and evaluation criteria.
- Evaluated end-to-end structure of the document intelligence system, benchmarking OCR accuracy, RAG retrieval quality, and routing performance, identifying key areas of improvement for production implementation.
- Improved document retrieval accuracy by 14% through tuning segmentation, indexing configuration, and model selection.
- Delivered a comprehensive technical report outlining model trade-offs, optimization strategies, and deployment recommendations, enabling stakeholders to make informed decisions on system rollout.

OPEN SOURCE CONTRIBUTIONS

Apache SQL Query Engine - apache / datafusion

Rust, SQL

- Reviewed and corrected unsafe optimizer behavior, preventing per-row traversal redundancy for reused query computations.
- Identified and helped resolve an execution issue ignoring DELETE predicates and removing unintended rows.
- Improved SQL validation by rejecting invalid user-defined function placeholders at definition time instead of runtime.

Datadog Observability Data Pipeline - vectordotdev / vector

Rust, Lua

- Increased Lua transform reliability by replacing unimplemented metric tag handling with structured errors.

Freebuff Coding Agent Harness - CodebuffAI / freebuff

TypeScript

- Fixed MCP tool argument loss by preserving complex schemas through model registration while maintaining runtime validation.

PROJECTS

GPU-Accelerated Postflop Poker Solver

C++, CUDA, FLTK

- Developed and deployed a custom No-Limit Hold'em postflop solver using Counterfactual Regret Minimization (CFR) across configurable game trees.
- Engineered flattened memory layouts and GPU kernels to maximize parallel execution, increasing iteration throughput by 7%.
- Applied memory-aware batching to support large workloads on consumer GPUs, reducing memory requirements by up to 60%.
- Profiled compute, memory-transfer, and allocation bottlenecks across the runtime, restructuring data access and execution paths to improve cache locality and reduce synchronization overhead.

GitHub Documentation RAG Service

Python, FastAPI, Qdrant, Docker, Google Cloud

- Built a modular RAG service that ingests public GitHub repositories and uses metadata-aware chunking, hybrid retrieval, and cross-encoder reranking to answer queries with grounded documentation and implementation specifications.
- Implemented configurable hybrid search combining dense embeddings, sparse retrieval, and cross-encoder reranking for complex documentation queries.
- Deployed typed API endpoints returning source-backed answers, validity, confidence, and completeness signals.
- Evaluated multiple indexing and ranking strategies, improving answer grounding to 92% faithfulness and 98% citation accuracy on a 50-question evaluation suite.

TECHNICAL SKILLS

Languages: Rust, C++, Python, Go, SQL

Systems: DataFusion, Arrow, Parquet, SQLite, MySQL, MongoDB, PostgreSQL, Qdrant, Redis

Performance: CUDA, multi-threading, profiling, vectorized execution

Infrastructure: Linux, Docker, Kubernetes, AWS, GCP